

## 4.5 Referential Integrity

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔

CS XI

Chapter 4: Data and Analysis

Lecture 07 of 15

★ Golden

### 4.5 Referential Integrity ★ Golden

#### LEARN IT

**Referential integrity** means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it. It stops **orphan records**.

If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student. You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows.

#### Parent — STUDENT

| ID  | Name |
|-----|------|
| 101 | Ali  |
| 102 | Sara |

#### Child — RESULT

| ID  | Grade        |
|-----|--------------|
| 101 | A            |
| 999 | F — rejected |

### Cascade operations

**Cascade Update.** Change the PK in the parent, and every matching FK in the child changes automatically. Ali's ID 101 → 201 updates Enrollments too. Nothing is left pointing at a vanished 101.

**Cascade Delete.** Delete the parent, and related child rows are deleted too. If Sara (102) leaves school, her Enrolment and Result rows disappear. That prevents "ghost" marks for a student who is gone.

#### WORKED EXAMPLE

##### Before Cascade Update

| Student.ID | Enrol.StudentID |
|------------|-----------------|
| 101        | 101             |

##### After changing 101 → 201 with Cascade Update ON

| Student.ID | Enrol.StudentID |
|------------|-----------------|
| 201        | 201             |

If Cascade Update were OFF, Access would refuse the change, or (in a sloppy file) leave Enrol stuck on 101 — an orphan.

#### COMMON MISTAKE

Cascade Delete is powerful and dangerous. Deleting one parent can wipe years of child history. In a real school, you more often **archive** a student (a Status field) than delete them.

#### EXAM TIP

Write: definition, why it matters (no orphans), then one numbered example of Cascade Update and one of Cascade Delete. Mention parent vs child.

#### اردو میں سمجھیں

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔ کیسکیڈ اپ ڈیٹ والدین کی بدلتے پر بچوں کو خود بدل دیتی ہے؛ کیسکیڈ ڈیلیٹ والدین مٹانے پر متعلقہ بچے ریکارڈ بھی مٹا دیتی ہے۔

#### CHECK YOURSELF

1. What is an orphan record?
2. Cascade Delete: parent Teacher is removed. What happens to that teacher's Class rows?

Answers: 1. A child row whose FK value does not exist in the parent. 2. They are deleted automatically (if Cascade Delete is on).

Next lecture: 4.6 Relational Schema Development